

Teacher Name: Zackery Duhala
 Subject(s): High School Biology
 Lesson # 1 of 3

Grade Level(s): 9/10th
 Title of Segment: DNA Structure & Variation
 Duration of Lesson: 120 minutes

Central Focus of Learning Segment:

The central focus of this learning segment is for students to develop and apply their understanding of DNA structure by analyzing nucleotide sequences to identify mutation types and explain how structural changes in DNA create genetic variation among individuals. Students engage in scientific practices by analyzing sequence data, identifying patterns, and constructing evidence-based explanations about how mutations contribute to variation. This foundation prepares students to later explain how DNA structure determines protein structure and cellular function.

Content Standards:

Students who demonstrate understanding can:

HS-LS1-1: Construct an explanation based on evidence for how the structure of DNA determines the structure of proteins which carry out the essential functions of life through systems of specialized cells. [Clarification Statement: N/A.] [Assessment Boundary: Assessment does not include identification of specific cell or tissue types, whole body systems, specific protein structures and functions, or the biochemistry of protein synthesis.]

Science and Engineering Practices	Disciplinary Core Ideas	Crosscutting Concepts
<u>Constructing Explanations and Designing Solutions</u> Constructing explanations and designing solutions in 9–12 builds on K–8 experiences and progresses to explanations and designs that are supported by multiple and independent student-generated sources of evidence consistent with scientific ideas, principles, and theories. Construct an explanation based on valid and reliable evidence obtained from a variety of sources (including students' own investigations, models, theories, simulations, peer review) and the assumption that theories and laws that describe the natural world operate today as they did in the past and will continue to do so in the future. (HS-LS1-1)	<u>LS1.A: Structure and Function</u> - Systems of specialized cells within organisms help them perform the essential functions of life. (HS-LS1-1) - All cells contain genetic information in the form of DNA molecules. Genes are regions in the DNA that contain the instructions that code for the formation of proteins, which carry out most of the work of cells. (HS-LS1-1) (Note: This Disciplinary Core Idea is also addressed by HS-LS3-1).	<u>Structure and Function</u> Investigating or designing new systems or structures requires a detailed examination of the properties of different materials, the structures of different components, and connections of components to reveal its function and/or solve a problem. (HS-LS1-1)
ELA/Literacy -		

Instructional Learning Objectives:	Assessments and Evidence of Learning:
By the end of this lesson, students will be able to describe the structural components of DNA and	Formative:
	Students complete a Think–Pair–Share using two nearly identical DNA sequences. They independently record similarities and differences, discuss with a partner, and share

explain how nucleotide sequence stores biological information. They will identify mutation types and analyze how structural changes alter that information, and construct a written explanation connecting DNA structure to its role in storing genetic information.	with the class. This formative assessment allows me to gauge students' understanding of DNA structure, base pairing, and the importance of nucleotide sequence order. By listening to their reasoning, I can identify misconceptions and adjust instruction before moving into mutation analysis. Summative: Students must explicitly connect DNA sequence to stored biological information and explain how mutation alters that information. Their responses demonstrate their ability to apply the crosscutting concept of Structure and Function and engage in the science and engineering practice of Constructing Explanations. The quality of students' written reasoning provides measurable evidence of their conceptual understanding and readiness to extend this knowledge to protein structure in subsequent lessons.
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Academic Language:	
Language Function:	
The primary language function for this lesson is to construct an evidence-based explanation. Students are required to analyze DNA sequences and explain how the structure of DNA determines the biological information it carries and how mutations alter that information. In order to meet this objective, students must move beyond identifying mutation types and instead articulate how changes in nucleotide order affect stored genetic information. This function aligns directly with HS-LS1-1 and the science and engineering practice of Constructing Explanations, as students must support their claims using evidence from sequence analysis.	
Vocabulary:	Additional Language Demands:
• Gene • Mutation • Substitution • Insertion • Deletion • Codon	Beyond vocabulary, students must use cause-and-effect language and structured scientific explanation to communicate their reasoning. Students are required to form complex sentences that link structural changes in DNA to changes in stored information. In addition, students must organize their written responses using scientific discourse conventions, including stating a clear claim, citing evidence from DNA sequence comparisons, and explaining how that evidence supports their claim. This requires students to synthesize data analysis with conceptual reasoning and communicate their thinking in a coherent and logical format.
Planned Language Supports:	
Modeling & Scaffolding: I will model one accurate DNA sequence comparison by identifying a substitution mutation and explaining how the change alters the order. I will also model one clear explanation statement. Language Supports: Students will use sentence frames such as “The mutation is a ____ because...,” and “This change alters the DNA structure by..., which changes the information because...”. Multiple Means of Representation: Directions will be read aloud and displayed visually.	

Anticipated Student Errors / Misconceptions:	
Some students may think that all mutations are harmful, or conversely, that a single base change cannot significantly affect biological information. Students may also confuse substitution, insertion, and deletion mutations. Additionally, students may forget or incorrectly	

apply base pairing rules (A–T and C–G) when completing complementary strands. These misconceptions will be addressed through modeling, guided questioning, and structured sequence comparisons during instruction.

Differentiation / Students Supports:

Short-Term English learners are permitted to use their phone or laptop to translate unfamiliar vocabulary to ensure comprehension of scientific terminology. Structured, scaffolded notes will be provided to guide students through key concepts and reduce cognitive load during instruction. I will explicitly model one mutation analysis problem step-by-step before students begin independent work to clarify expectations and reasoning processes. Students are allowed to work in partners during sequence comparisons. Additionally, one of my IEP students is permitted to listen to music with headphones to support focus and task persistence.

Grouping and Materials:

Grouping:

Students will engage in a variety of grouping structures throughout the lesson to support collaboration. The lesson will begin with whole-class instruction during the Engage and initial modeling phases. During the Explore phase, students will work in previously made pairs to compare DNA sequences and identify mutation types before writing independently. These pairs are designed to benefit the students' individual learning needs (IEP student in front, EL learners together, gifted students together).

Materials:

- “DNA Variation Slides”
- “Mutation Lecture Notes”
- “Mutations Worksheet”
- “Mutations Vocab Quiz-google form”
- Laptops
- Colored pencils for mutations
- Devices used for translation of vocab

Engage:

Teacher Does:

The teacher displays two nearly identical DNA sequences on the board, highlighting one small difference between them. The teacher asks students guiding questions such as, “What do you notice about these two sequences?” and “If only one base changed, does that matter?” The learning objective is then introduced, emphasizing that DNA structure stores biological information and that small changes in structure may alter that information.

Student Does:

Students independently observe the displayed DNA sequences and write down what they notice about similarities and differences. They then participate in a Think–Pair–Share discussion with a partner, explaining their observations and predicting whether the small structural change might affect genetic information. Students share responses with the class.

Check for understanding: How will I check for understanding at this stage?

The teacher listens to student responses during the Think–Pair–Share and asks targeted follow-up questions such as, “What specifically changed in the sequence?” and “Why might the order of bases matter?” I will call on multiple students to explain their reasoning using academic vocabulary and observe whether they correctly identify structural differences between the sequences. If students demonstrate confusion about sequence order or the significance of structural changes, I will clarify before moving into the Explore phase.

Explore:

Teacher Does:

Student Does:

The teacher distributes Mutation Worksheets that include original and mutated sequences. I first model one example step-by-step, demonstrating how to align sequences, apply base pairing rules, and identify the type of mutation. After modeling, I circulate the room as students work in pairs, asking probing questions. I provide clarification, highlight key vocabulary, and offer additional scaffolds as needed.	Students work in partners to compare original and mutated DNA sequences. They align the sequences, circle differences, identify the mutation type, and explain how the structural change alters the bases order. They apply base pairing rules when completing complementary strands, and using the lesson vocabulary and sentence frames as needed.
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Check for understanding: *How will I check for understanding at this stage?*

The teacher assesses understanding by circulating and reviewing students' sequence comparisons as they work in pairs. Asking targeted questions to determine whether students can accurately identify mutation types and explain structural differences. I will listen for correct use of vocabulary and observe whether students are correctly aligning sequences and applying base pairing rules. If students demonstrate misconceptions or confusion, I will pause to provide clarification or re-model the process before moving forward.

Explain:

Teacher Does:

I explicitly connect students' sequence analysis to the concept that the structure (order) of DNA determines the information it carries. Using the projected model, I highlight how even a small structural change in a sequence order alters the overall message stored in the sequence.

Student Does:

Students share their identified mutation examples and explain their reasoning using scientific vocabulary. They practice forming complete explanations verbally before preparing to write independently in the next phase.

Elaborate:

Teacher Does:

Provides additional mutation examples and prompts students to compare the impact of different structural changes.

Student Does:

Analyze new sequences, determine which mutations alter more of the structure, and write a brief explanation linking structure to information change.

Evaluate:

Teacher Does:

Assigns a written response asking students to explain how DNA structure determines information and how mutations change it.

Student Does:

Independently construct an evidence-based explanation using correct vocabulary and cause-and-effect reasoning.

Check for understanding: *How will I check for understanding at this stage?*

The teacher will review students' written explanations to ensure they correctly connect DNA structure to stored genetic information and clearly explain how a mutation alters that information.

Closure:

Students will complete a brief vocabulary quiz covering key terms. After the quiz, I will briefly reinforce the connection between DNA structure and stored information and preview the next lesson, where we will build on this foundation.

Resources:

-  DNA Variation Slides
-  Unit 7: Mutations Lecture & Worksheet

Teacher Name: Zackery Duhala
 Subject(s): High School Biology
 Lesson # 2 of 3

Grade Level(s): 9/10th
 Title of Segment: DNA Forensics Activity
 Duration of Lesson: 120 minutes

Central Focus of Learning Segment:

The central focus of this lesson is for students to apply their understanding of DNA structure and mutation to analyze forensic DNA evidence and construct an evidence-based explanation identifying which suspect most closely matches the crime scene DNA sample. Students use DNA sequences as models of stored biological information and justify their conclusions using scientific reasoning.

Content Standards:

Students who demonstrate understanding can:

HS-LS1-1: Construct an explanation based on evidence for how the structure of DNA determines the structure of proteins which carry out the essential functions of life through systems of specialized cells: [Clarification Statement: N/A.] [Assessment Boundary: Assessment does not include identification of specific cell or tissue types, whole body systems, specific protein structures and functions, or the biochemistry of protein synthesis.]

Science and Engineering Practices	Disciplinary Core Ideas	Crosscutting Concepts
<u>Constructing Explanations and Designing Solutions</u> Constructing explanations and designing solutions in 9–12 builds on K–8 experiences and progresses to explanations and designs that are supported by multiple and independent student-generated sources of evidence consistent with scientific ideas, principles, and theories. Construct an explanation based on valid and reliable evidence obtained from a variety of sources (including students' own investigations, models, theories, simulations, peer review) and the assumption that theories and laws that describe the natural world operate today as they did in the past and will continue to do so in the future. (HS-LS1-1)	<u>LS1.A: Structure and Function</u> - Systems of specialized cells within organisms help them perform the essential functions of life. (HS-LS1-1) - All cells contain genetic information in the form of DNA molecules. Genes are regions in the DNA that contain the instructions that code for the formation of proteins, which carry out most of the work of cells. (HS-LS1-1) (Note: This Disciplinary Core Idea is also addressed by HS-LS3-1).	<u>Patterns</u> Different patterns may be observed at each of the scales at which a system is studied and can provide evidence for causality in explanations of phenomena.

ELA/Literacy -

Instructional Learning Objectives:	Assessments and Evidence of Learning:
By the end of this lesson, students will be able to systematically compare DNA sequences, accurately count	Formative: The teacher will observe during sequence comparison and mismatch counting. The teacher will monitor students' ability to align sequences correctly and quantify differences accurately. Students will complete a structured data table ranking suspects

mismatches, and organize their findings to determine which suspect's DNA most closely matches the crime scene sample. Students will use structural patterns in DNA as evidence for future argument construction.	by similarity. The completed comparison chart will serve as the lesson-level assessment, demonstrating accurate data collection and analysis in preparation for argument construction in Lesson 3.
	Summative: The summative assessment for this lesson is the completed suspect DNA table. Students will systematically compare the crime scene DNA sequence with each suspect's sample, identify structural differences, accurately count mismatches, and rank suspects based on sequence similarity. Students will include a brief written justification explaining which suspect currently appears most similar based on pattern analysis. This assessment measures students' ability to analyze DNA structure, recognize patterns in sequence data, and organize evidence in preparation for formal argument construction in Lesson 3.

Academic Language: Students must use vocabulary including DNA sample, sequence, mismatch, mutation, similarity, and pattern. Students must describe structural differences using precise comparative language such as “more similar,” “fewer mismatches,” and “greater structural change.”	
Language Function: Students must systematically examine DNA patterns, identify structural differences between samples, and describe levels of similarity using precise scientific vocabulary. This prepares students to use their analyzed evidence to construct a formal scientific argument in the following lesson.	
Vocabulary: • Mismatch • Forensic	Additional Language Demands: In this lesson, students must use comparative and quantitative language to describe patterns in DNA sequence data. They are required to explain similarities and differences using phrases such as “fewer mismatches,” “more similar,” and “greater structural difference.” Students must also organize their thinking when recording data in tables and ranking suspects by similarity. In addition, students must justify their rankings using clear cause-and-effect reasoning, explaining how the number of mismatches reflects structural similarity between DNA samples.
Planned Language Supports: The teacher will model how to use comparative language and quantitative evidence when explaining similarity between DNA samples. A vocab sheet will be given to students, and can be used throughout the lesson. Students with English learning accommodations are allowed to use devices for translation throughout the activity.	

Anticipated Student Errors / Misconceptions: Students may compare sequences visually without systematically counting mismatches, misalign the second strand of DNA, or assume that one mismatch automatically eliminates a suspect without deeper analysis. These issues will be addressed through modeling and structured data organization.

Differentiation / Students Supports: English learners may use translation tools for vocabulary clarification. Structured data tables will scaffold systematic comparison. The teacher will model one full example of mismatch

counting before independent work. Students may work in partners to support collaborative reasoning. IEP students may use headphones for focus, per their accommodation. Advanced students may attempt extension questions for extra credit.

Grouping and Materials:

Grouping:	Materials:
Instruction will begin with whole-class modeling before transitioning to a group setting. Groups will be sized 3-4, similar to their partner groups they will be grouped with students in their range. STEL's must be paired with other spanish speakers per school code. Students will complete structured comparison charts.	• DNA Variation Slides • Forensics Activity Worksheet (version A, B, or C) • Suspect DNA printout (available at stations)

Engage:

Teacher Does:	Student Does:
The teacher will reintroduce the forensic scenario and ask students how scientists determine DNA matches, and ask how DNA acts as a source to identify individuals.	Students will briefly discuss how patterns in sequence structure might reveal similarity. Students discuss how DNA is individualized.

Check for understanding: *How will I check for understanding at this stage?*

The teacher will use multiple ways of CFU, such as volunteers, group agreement or disagreement with hand signals, and the use of equity stick (flippity app) to have students repeat learning.

Explore:

Teacher Does:	Student Does:
The teacher will model how to align sequences and count mismatches step-by-step, between crime scene DNA and "suspect" samples.	Students will then compare the crime scene DNA with each suspect sample, mark structural differences, count mismatches, and record results in a comparison table. The teacher will circulate to ensure accurate alignment and reasoning, engaging with students who need assistance in analysis or comparison.

Check for understanding: *How will I check for understanding at this stage?*

The teacher will assess student understanding by circulating and reviewing how students align DNA sequences and count mismatches. Targeted questions will be asked to ensure students are using a systematic comparison method rather than guessing. The teacher will listen for accurate use of vocabulary such as sequence, mismatch, and similarity, and confirm that students are correctly recording their data in the comparison chart before moving to the next suspect.

Explain:

Teacher Does:	Student Does:
The teacher will facilitate group to group discussion. They will guide students to articulate how structural similarity is determined by consistent DNA sequence patterns and reinforce the use of precise	Students share their findings, explain how they counted mismatches, and describe which suspect appears most similar based on structural patterns. They use comparative language such as "fewer mismatches" or

vocabulary such as sequence, mismatch, similarity, and pattern.	“greater similarity” to justify their reasoning. Students listen to peers’ explanations and revise their own rankings if necessary.
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Elaborate:

Teacher Does:	Student Does:
The teacher will challenge students to think more critically about the reliability of their analysis by posing questions such as, “What if two suspects had the same number of mismatches?” and “Would one mismatch automatically eliminate a suspect?” The teacher prompts students to consider how additional evidence or longer DNA sequences might increase confidence in a conclusion. The teacher guides students to recognize that stronger pattern consistency across a sequence increases reliability in forensic identification.	Students complete analysis questions. They may revisit their data table to reassess similarity and consider how more sequence information could impact their ranking. Students use comparative and analytical language to explain how pattern consistency affects certainty. Extension higher level questions are available to attempt.

Evaluate:

Teacher Does:	Student Does:
The teacher distributes stamps to students with the correct “suspect”. The teacher will review their mismatch counts, sequence alignment accuracy, and logical ranking of suspects based on structural similarity. The teacher will ensure that students have used consistent methods to analyze the data and have recorded their findings clearly in the structured table.	Students will save their Forensics activity worksheet, once they have earned their stamp. This worksheet will be used to complete the CER in lesson 3.

Check for understanding: *How will I check for understanding at this stage?*

The teacher will mark the seating chart roster, and make note of the students who have successfully found the correct match to their crime scene DNA.

Closure:

Students will summarize how analyzing patterns in DNA structure helped narrow down suspects. The teacher will preview Lesson 3, explaining that students will use this collected evidence to construct a formal Claim-Evidence-Reasoning argument identifying the matching DNA sample.

Resources:

- ☐ DNA Variation Slides
- ☐ Forensics Activity Worksheet A
- ☐ Forensics Activity Worksheet B
- ☐ Forensics Activity Worksheet C
- ☐ Suspect DNA Strands

Teacher Name: Zackery Duhala
 Subject(s): High School Biology
 Lesson # 3 of 3

Grade Level(s): 9/10th
 Title of Segment: Forensic Assessment
 Duration of Lesson: 50 minutes

Central Focus of Learning Segment:

The central focus of this lesson is for students to construct a scientific argument using evidence from DNA sequence analysis to determine which suspect's DNA matches the crime scene sample. Students synthesize their data from Lesson 2 and apply their understanding of DNA structure to explain how structural similarity in DNA sequences supports identification. This lesson emphasizes constructing evidence-based explanations.

Content Standards:

Students who demonstrate understanding can:

HS-LS1-1: Construct an explanation based on evidence for how the structure of DNA determines the structure of proteins which carry out the essential functions of life through systems of specialized cells. [Clarification Statement: N/A.] [Assessment Boundary: Assessment does not include identification of specific cell or tissue types, whole body systems, specific protein structures and functions, or the biochemistry of protein synthesis.]

Science and Engineering Practices	Disciplinary Core Ideas	Crosscutting Concepts
<u>Constructing Explanations and Designing Solutions</u> Constructing explanations and designing solutions in 9–12 builds on K–8 experiences and progresses to explanations and designs that are supported by multiple and independent student-generated sources of evidence consistent with scientific ideas, principles, and theories. Construct an explanation based on valid and reliable evidence obtained from a variety of sources (including students' own investigations, models, theories, simulations, peer review) and the assumption that theories and laws that describe the natural world operate today as they did in the past and will continue to do so in the future. (HS-LS1-1)	<u>LS1.A: Structure and Function</u> - Systems of specialized cells within organisms help them perform the essential functions of life. (HS-LS1-1) - All cells contain genetic information in the form of DNA molecules. Genes are regions in the DNA that contain the instructions that code for the formation of proteins, which carry out most of the work of cells. (HS-LS1-1) (Note: This Disciplinary Core Idea is also addressed by HS-LS3-1).	<u>Patterns</u> Different patterns may be observed at each of the scales at which a system is studied and can provide evidence for causality in explanations of phenomena.

ELA/Literacy -

Instructional Learning Objectives:	Assessments and Evidence of Learning:
By the end of this lesson, students will construct a complete Claim-Evidence-Reaso	Formative: Formative assessment during this lesson occurs as students draft their CER responses and participate in peer review. The teacher will circulate while students write, checking for a clear claim,

ning (CER) argument identifying which suspect most likely left the DNA sample. Students will use data as evidence and explain how structural similarity in DNA supports their conclusion.	accurate use of data as evidence, and logical reasoning that connects DNA structural similarity to their conclusion. The teacher will ask students to verbally explain how their evidence supports their claim before submission. Feedback using the rubric will also serve as formative assessment, allowing students to revise and strengthen their arguments prior to final submission.
	Summative: Students must state a clear claim identifying the suspect, provide at least two pieces of numerical evidence (mismatch counts), and explain how structural similarity in DNA supports their claim. The rubric included in the CER document will guide evaluation. This assessment measures students' ability to synthesize data analysis and construct an evidence-based scientific argument.

Academic Language: The language function for this lesson is to construct an argument supported by evidence. Students must use precise vocabulary including claim, evidence, reasoning, sequence, mismatch, similarity, mutation, and DNA sample. Students must organize their writing logically by stating a claim, citing quantitative evidence, and explaining how that evidence supports the conclusion. Cause-and-effect reasoning and comparative language are required.

Language Function:

The primary language function for this lesson is to construct an argument supported by evidence. Students must develop a clear claim identifying the suspect, provide DNA mismatch data as evidence, and explain how that evidence supports their conclusion.

Vocabulary:

- Claim
- Evidence
- Reasoning

Additional Language Demands:

Students must organize their writing using the structured CER format, clearly separating claim, evidence, and reasoning. They are required to use comparative and quantitative language and employ cause-and-effect reasoning to explain how DNA structural similarity supports their conclusion.

Planned Language Supports:

To support students in meeting these language demands, the teacher will model a complete CER example and explicitly identify each component (claim, evidence, reasoning). Students will use the structured CER template provided on the worksheet and sentence frames such as "I claim that...", "One piece of evidence is...", and "This evidence supports my claim because...". Students will rehearse their reasoning verbally with a partner before finalizing their written response.

Anticipated Student Errors / Misconceptions:

Students may restate evidence without explaining how it supports the claim, confuse evidence with reasoning, or make claims that are not fully supported by their data. Some students may rely on general statements rather than specific evidence. These issues will be addressed through modeling and guided revision.

Differentiation / Students Supports:

Students will use the structured CER template provided in the forensic worksheet. The teacher will model one strong example of a CER claim before independent writing. Sentence frames such as “I claim that...,” “One piece of evidence is...,” and “This evidence supports my claim because...” will scaffold writing. English learners may use translation tools for vocabulary clarification, as well as be offered the option of a paper or online copy of the CER (allowing built-in translation accommodation). Students may ask the teacher for clarification and assistance if needed. IEP students may use headphones to maintain focus.

Grouping and Materials:

Grouping:	Materials:
Instruction will begin with whole-class modeling of a CER structure. Students will then work independently on their written argument, with optional teacher guidance to submission.	• DNA Variation L Sides • Forensics Activity data • Forensic Activity CER • Sentence Starters • CER Rubric

Engage:

Teacher Does:	Student Does:
The teacher will review the forensic scenario and ask students, “Based on your data from yesterday, who do you think matches the crime scene DNA?” Students will briefly discuss their preliminary conclusions	Students will discuss findings, and prepare thoughts for CER completion.

Check for understanding: *How will I check for understanding at this stage?*

The teacher will listen for accurate references to mismatch counts and confirm that students understand that structural similarity is determined by sequence patterns before transitioning to formal argument writing.

Explore:

Teacher Does:	Student Does:
Introduce the CER structure using the forensic worksheet and model one complete claim, clearly identifying the claim, possible evidence, and reasoning components. The teacher will then demonstrate how to integrate data into a coherent explanation.	Students identify the claim, evidence, and reasoning in the modeled example and begin drafting their own CER using their collected data.

Check for understanding: *How will I check for understanding at this stage?*

The teacher will ask students to explain verbally which part of their draft represents the claim and which part represents evidence, ensuring they understand the structural components of scientific argumentation before writing independently.

Explain:

Teacher Does:	Student Does:
The teacher will circulate while students draft their CER responses, providing feedback on clarity of claims, accuracy of evidence, and completeness of reasoning. Prompting students to move beyond listing data and explain how their evidence supports their conclusion.	Students refine their CER responses, incorporating vocabulary from the unit.

Elaborate:

Teacher Does:	Student Does:
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The teacher guides students to check for clear claims, sufficient evidence, and logical reasoning.	Students revise their writing to strengthen clarity and justification.
Evaluate:	
Teacher Does:	Student Does:
The teacher will collect students' final CER responses and assess them using the rubric criteria.	Students submit their completed CER, demonstrating their ability to construct an evidence-based argument supported by DNA sequence patterns.
Check for understanding: <i>How will I check for understanding at this stage?</i>	
The teacher will review whether students clearly connect DNA structural similarity to their conclusion, confirming mastery of argument construction aligned with the DNA Structure and pattern element of HS-LS1-1.	

Closure:
To conclude the lesson, the teacher will ask students to reflect on how analyzing patterns in DNA structure allowed them to identify a matching suspect. I will prompt them with the question, "How did the structure of DNA help you justify your conclusion?" Students will briefly share responses, reinforcing the connection between DNA sequence patterns and evidence-based reasoning. Then will then connect this forensic application back to the broader scientific idea that DNA structure stores biological information, emphasizing that scientific conclusions must be supported by clear structural evidence.

Resources:
 DNA Variation Slides  Forensics Activity CER & Rubric